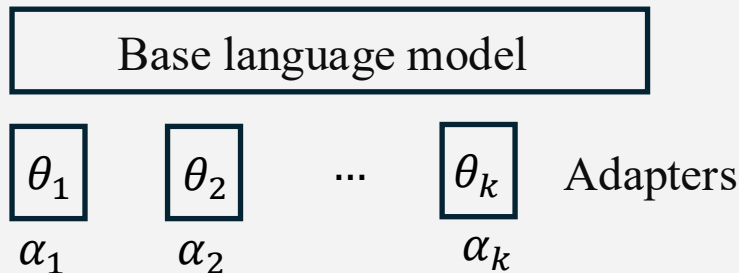


Input: a dataset of n rankings, each over m responses to an input prompt

$$\left\{ \begin{array}{l} \text{Input prompt: } x_i \\ \text{Responses: } Y_i: \{y_{i,1}, \dots, y_{i,m}\} \\ \text{Ranking: } y_{i,\sigma_i(1)} \succ \dots \succ y_{i,\sigma_i(m)} \end{array} \right\}_{i=1}^n$$

Require: A base model π_{ref} , output function r_θ , and input embeddings $h(x, y)$

Output: A mixture of k Plackett-Luce models, as k adapters on a base language model

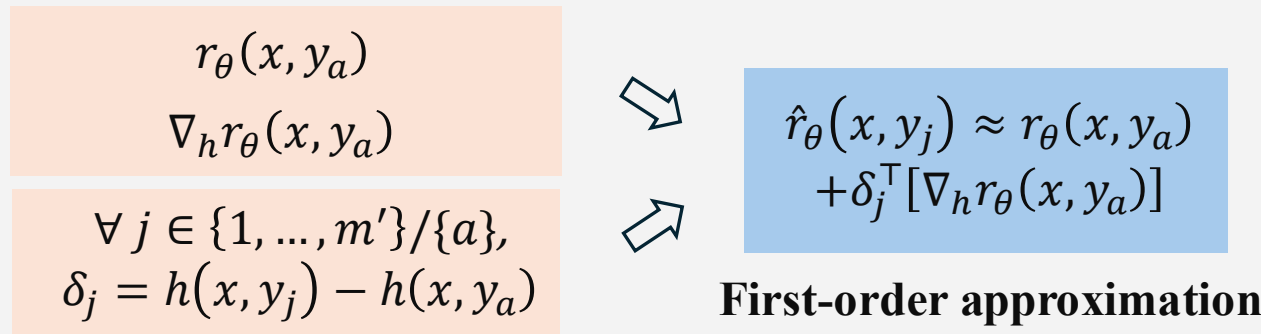


Augmenting the ranking size before training



Output a partial ranking: $y_{\sigma_i(1)} \succ \dots \succ y_{\sigma_i(m)} \succ \{y_{m+1}, \dots, y_{m'}\}$

Efficiently learning Plackett-Luce models via gradient estimation



Estimate the ranking probability $\hat{P}_\theta(\sigma|x, Y')$ during training:

$$\hat{P}_\theta(\sigma|x, Y') = \prod_{t=1}^m \frac{\exp(\hat{r}_\theta(x, y_{\sigma(t)}))}{\sum_{j=t}^m \exp(\hat{r}_\theta(x, y_j)) + \sum_{j=m+1}^{m'} \exp(\hat{r}_\theta(x, y_j))}$$